The use of IEEE Standard 738-2023 is experimental and not yet intended for production use.

Transient Thermal or Fault Rating

IEEE Standard 738-2023 method of calculation

Weather Conditions

```
Air temperature:                40.000 (deg C)
Wind speed:                      0.610 (m/s)
Wind to conductor angle:        90.000 (deg)
Atmosphere type:            Industrial
```

Solar Conditions

```
Date:  June 21 (day of the year with most solar heating)
Day of year:                                172
Sun time:                                11.000 (hours)
Altitude:                                76.024 (deg)
Azimuth:                                100.546 (deg)
```

Conductor Properties

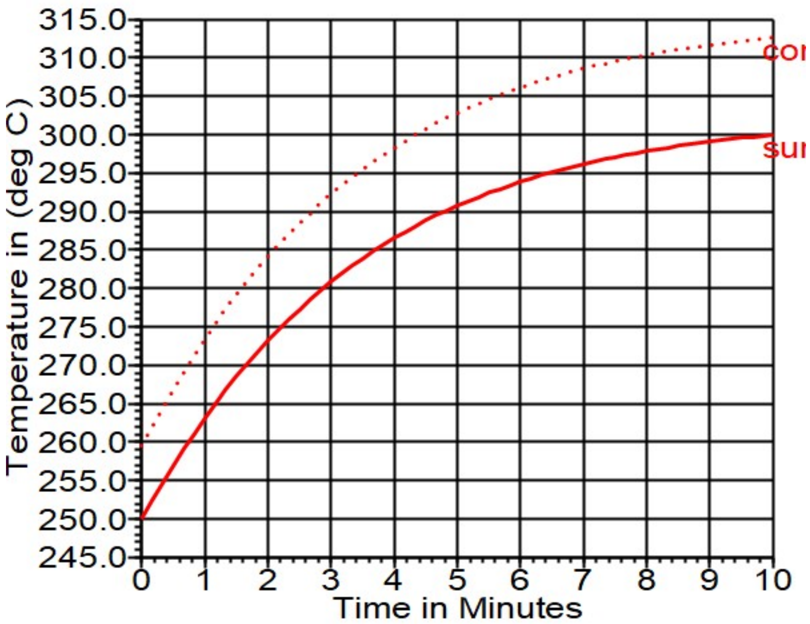
```
Description: 155.4 kcmil 7/0 Strands ANAHEIM AAAC - Adapted from 1970's Publicly Available Data
Azimuth:                           90.000 (deg)
AC resistance at 25.0 (deg C):    0.4330 (Ohm/km)
AC resistance at 75.0 (deg C):    0.5068 (Ohm/km)
Solar absorptivity:                 0.500
Emissivity:                         0.500
Outer diameter:                    11.354 (mm)
Outer strand diameter:              0.000 (mm)
Outer strand layers:                    0
Outer surface finish:              Smooth
Cable is thermal bimetallic:        False
Heat capacity:                    205.849 (Watt-s/m-deg C)
Maximum allowable temperature:    300.000 (deg C)
```

```
Latitude:                          26.800 (deg)
Elevation above sea level:        100.000 (m)

Analysis Results
----------------
Initial steady current:                559.900 (Amps)
Initial steady temperature:            249.924 (deg C)
Total analysis time:                    10.000 (Min)
Analysis time interval:                 10.000 (Sec)
Step current (applied at T=0.0):       615.657 (Amps)
Surface temperature at end time:       300.000 (deg C)
Final conductor azimuth:                90.000 (deg)
Final solar absorptivity:                0.500
Final emissivity:                        0.500
Final wind to conductor angle:          90.000 (deg)

Time=    0.0000 sec   Conductor temperature=  249.9 (deg C)
Time=   10.0000 sec   Conductor temperature=  252.4 (deg C)
Time=   20.0000 sec   Conductor temperature=  254.7 (deg C)
Time=   30.0000 sec   Conductor temperature=  257.0 (deg C)
Time=   40.0000 sec   Conductor temperature=  259.1 (deg C)
Time=   50.0000 sec   Conductor temperature=  261.2 (deg C)
Time=   60.0000 sec   Conductor temperature=  263.2 (deg C)
Time=   70.0000 sec   Conductor temperature=  265.1 (deg C)
Time=   80.0000 sec   Conductor temperature=  266.9 (deg C)
Time=   90.0000 sec   Conductor temperature=  268.6 (deg C)
Time=  100.0000 sec   Conductor temperature=  270.2 (deg C)
Time=  110.0000 sec   Conductor temperature=  271.8 (deg C)
Time=  120.0000 sec   Conductor temperature=  273.3 (deg C)
Time=  130.0000 sec   Conductor temperature=  274.7 (deg C)
Time=  140.0000 sec   Conductor temperature=  276.1 (deg C)
Time=  150.0000 sec   Conductor temperature=  277.3 (deg C)
Time=  160.0000 sec   Conductor temperature=  278.6 (deg C)
Time=  170.0000 sec   Conductor temperature=  279.8 (deg C)
Time=  180.0000 sec   Conductor temperature=  280.9 (deg C)
Time=  190.0000 sec   Conductor temperature=  281.9 (deg C)
Time=  200.0000 sec   Conductor temperature=  283.0 (deg C)
Time=  210.0000 sec   Conductor temperature=  283.9 (deg C)
Time=  220.0000 sec   Conductor temperature=  284.9 (deg C)
Time=  230.0000 sec   Conductor temperature=  285.7 (deg C)
Time=  240.0000 sec   Conductor temperature=  286.6 (deg C)
Time=  250.0000 sec   Conductor temperature=  287.4 (deg C)
Time=  260.0000 sec   Conductor temperature=  288.1 (deg C)
Time=  270.0000 sec   Conductor temperature=  288.8 (deg C)
```

```
Time=  280.0000 sec   Conductor temperature=  289.5 (deg C)
Time=  290.0000 sec   Conductor temperature=  290.2 (deg C)
Time=  300.0000 sec   Conductor temperature=  290.8 (deg C)
Time=  310.0000 sec   Conductor temperature=  291.4 (deg C)
Time=  320.0000 sec   Conductor temperature=  291.9 (deg C)
Time=  330.0000 sec   Conductor temperature=  292.5 (deg C)
Time=  340.0000 sec   Conductor temperature=  293.0 (deg C)
Time=  350.0000 sec   Conductor temperature=  293.4 (deg C)
Time=  360.0000 sec   Conductor temperature=  293.9 (deg C)
Time=  370.0000 sec   Conductor temperature=  294.3 (deg C)
Time=  380.0000 sec   Conductor temperature=  294.7 (deg C)
Time=  390.0000 sec   Conductor temperature=  295.1 (deg C)
Time=  400.0000 sec   Conductor temperature=  295.5 (deg C)
Time=  410.0000 sec   Conductor temperature=  295.9 (deg C)
Time=  420.0000 sec   Conductor temperature=  296.2 (deg C)
Time=  430.0000 sec   Conductor temperature=  296.5 (deg C)
Time=  440.0000 sec   Conductor temperature=  296.8 (deg C)
Time=  450.0000 sec   Conductor temperature=  297.1 (deg C)
Time=  460.0000 sec   Conductor temperature=  297.4 (deg C)
Time=  470.0000 sec   Conductor temperature=  297.6 (deg C)
Time=  480.0000 sec   Conductor temperature=  297.9 (deg C)
Time=  490.0000 sec   Conductor temperature=  298.1 (deg C)
Time=  500.0000 sec   Conductor temperature=  298.3 (deg C)
Time=  510.0000 sec   Conductor temperature=  298.5 (deg C)
Time=  520.0000 sec   Conductor temperature=  298.7 (deg C)
Time=  530.0000 sec   Conductor temperature=  298.9 (deg C)
Time=  540.0000 sec   Conductor temperature=  299.1 (deg C)
Time=  550.0000 sec   Conductor temperature=  299.3 (deg C)
Time=  560.0000 sec   Conductor temperature=  299.4 (deg C)
Time=  570.0000 sec   Conductor temperature=  299.6 (deg C)
Time=  580.0000 sec   Conductor temperature=  299.7 (deg C)
Time=  590.0000 sec   Conductor temperature=  299.9 (deg C)
Time=  600.0000 sec   Conductor temperature=  300.0 (deg C)
```


Temperature in (deg C)
315.0
310.0
305.0
300.0
295.0
290.0
285.0
280.0
275.0
270.0
265.0
260.0
255.0
250.0
245.0
0
1
2
3
4
5
6
7
8
9
10
Time in Minutes
co
su